Bio-Unix & Shell Scripting

The Linux file system

prepared by

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Introduction to file system hierarchy

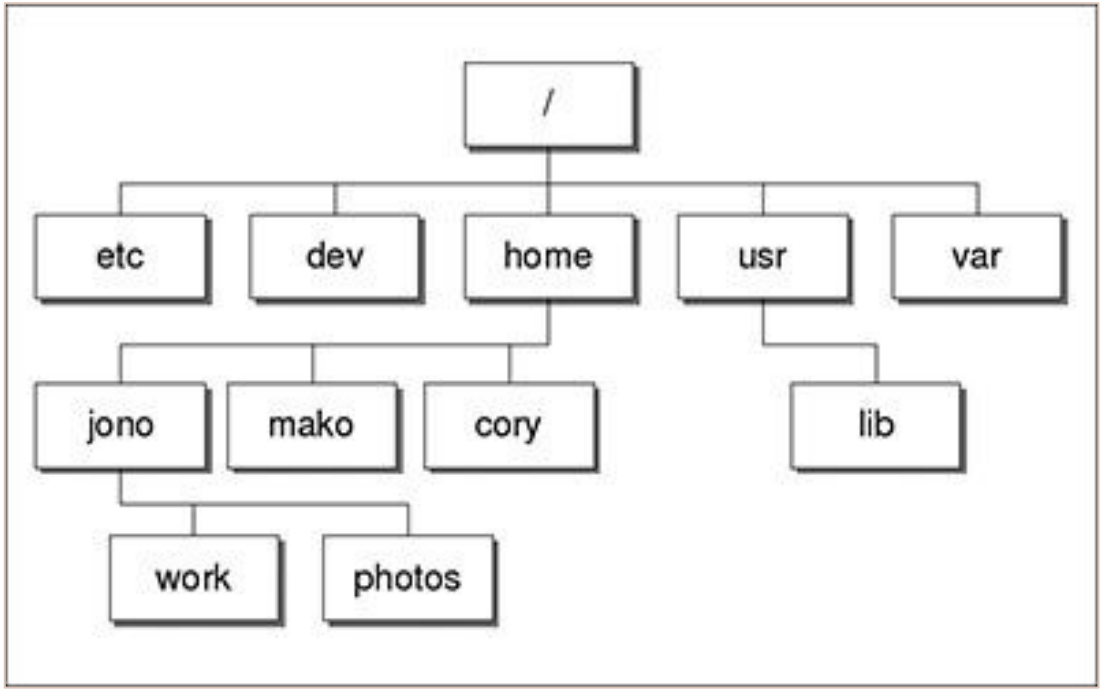
- Unix-like operating system such as Linux and its distributions have a hierarchical directory structure, just like in Windows OS
- This means the directories are organised in a tree-like pattern and each directory may contain contain files and other directories
- A directory is usually referred to as a folder in Windows Operating Systems

Overview of directory categories

- Systems binaries; that harbor core commands
- Boot and kernel-related directories; that help the system start
- Configuration files; that define how Linux behaves
- User related (personal) directories
- Shared Libraries; that support essential programs
- System and run time information; that reflects what is happening in real time
- Mount points and media (External devices)
- Multi-user resources; for software and locks
- Temporary files; used briefly by the system
- Optional software: that is installed separately

File system hierarchy

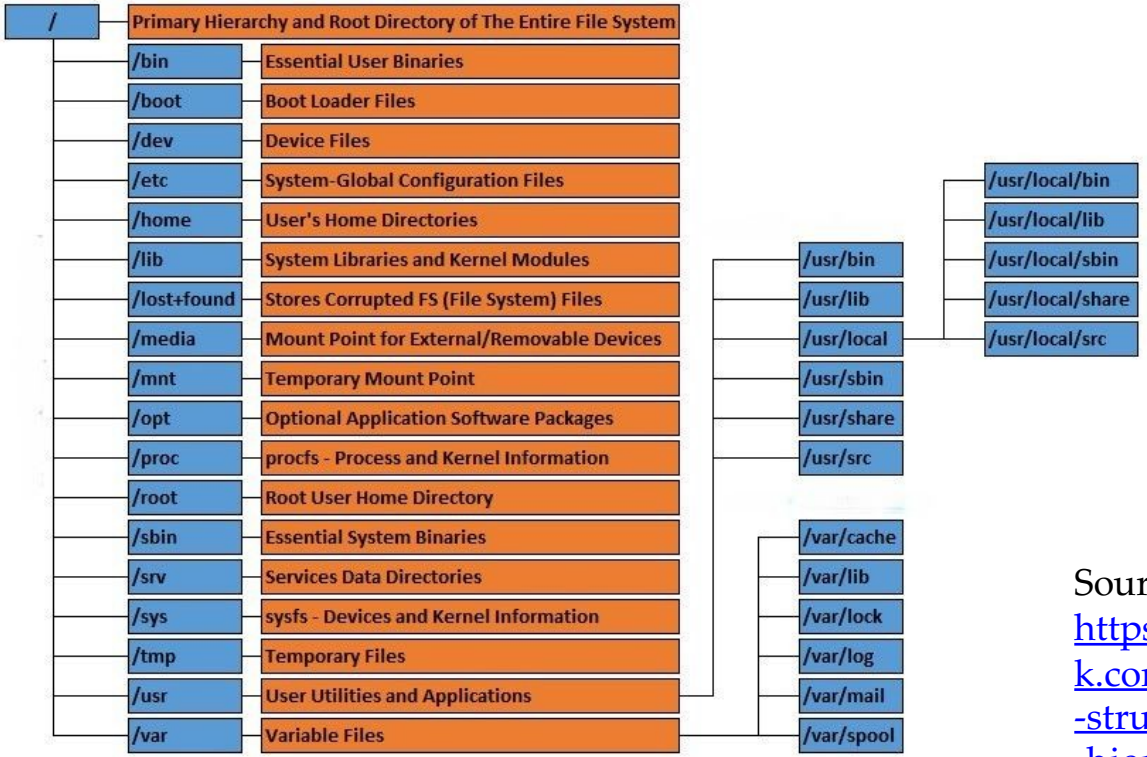
- This file system hierarchy is common in most linux distributions with minor differences



File system hierarchy

- /: The root directory is the first directory and stores all the other directories
- /bin: Contains essential user binaries. Contains common command line user commands and other binary executable files
- /boot: Contains all the files required for the boot process
- /home: Contains the users' directories where the users can create and manipulate files
- /mnt: Mount points for removable storage devices

File system hierarchy - detailed



Source: https://www.2daygeek.com/linux-directory-structure-file-system-hierarchy/

Linux Command Syntax

- Command -[options] [arguments]
- Command --[options] [arguments]
- Options modify the behavior of the command whereas the arguments are the items upon which the command acts e.g files, directories, and data
- options are preceded by a hyphen (-) when written in short format or (--) when written in their full form

Navigating the file system

Commands

- *pwd* - Print name of current working directory
- *ls* - List directory contents
- *cd* - Change directory

Navigating the file system - *pwd*

- At any one point, a user is within a given directory and may have directories above or subdirectories below
- The directory you are currently in is called the current working directory
- When you launched the terminal for the first time, the home directory will be your current working directory
- Every user account has its own home directory where a regular user is allowed to write files

Navigating the file system - *ls*

- *ls* command will list the contents of your current working directory or any other directory of your choice

Common *ls* options

- *ls -l* display output in a long format
- *ls -1* lists one file/directory per line
- *ls -r* display contents in reverse order

Navigating the file system - *ls*

Common *ls* options

- *ls -a* lists all the contents of a given directory
- *ls -h* display file size in human readable format
- *-ls -t* display contents by modification time

- Note: To explore other options type *ls - - help* to show the help page or *man ls* to show the manual page. *man* and *help* can be used with most commands

Navigating the file system - *cd*

- The *cd* command enables a user to move from the current directory into another directory
- The cd command is typed and followed by the path to the desired directory
- the path is the route you take to get to your desired directory and may be absolute or relative
- If a path is not specified, this command will default to the home directory of the user which can also be detonated by the tilde (~)

Navigating the file system - *cd*

Absolute path

- The absolute path starts from the root (/) directory upto the desired directory or file e.g *cd /usr/bin/*

Relative path

- The relative path starts from your current working directory upto the desired directory or file. e.g *cd bash2023/malaria/file.txt* assuming bash2023 is my current working directory

Navigating the file system - *cd*

- When using the *cd* command, the notation . also refers to the current working directory and .. refers to the parent directory of current working directory
- *cd -* will change directory to the previous directory

Thank you.